Practice workbook · Explain how the pupil responds to changes in light intensity

Iris muscle actions that widen or narrow the pupil in dim or bright light and why a quick response protects the retina.

4 question parts: 1 worked-example retry and 3 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

OLevels.com SCHOOL

Type ID: HBI-EXPLAIN-PUPIL-LIGHT-RESPONSE

1. Question 6 · part b.i · 5 marks

June 2021 Paper 2

Worked-example retry · Attempt part b.i only.

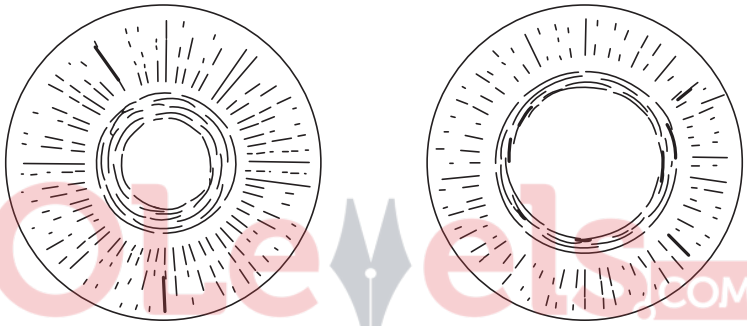
6 The human eye consists of many parts.

(a) Which part has the highest concentration of light sensitive cells?

(1)

- ☒ **A** blind spot
- ☒ **B** choroid
- ☒ **C** fovea
- ☒ **D** optic nerve

(b) The diagram shows the iris of an eye in two different light conditions, A and B.



light condition A

light condition B

(i) Describe what happens in the eye when the light condition changes from A to B.

(5)

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

P 6 6 4 3 3 A 0 1 4 2 4

1. Question 6 · part b.i · 5 marks

June 2021 Paper 2

Worked-example retry · Attempt part b.i only.

DO NOT WRITE IN THIS AREA

(ii) Explain why it is important that the changes in the eye occur quickly when the light condition changes from B to A.

(3)

(Total for Question 6 = 9 marks)

OLevels.com SCHOOL

Working and answer

2. Question 6 · part b.ii · 3 marks

June 2021 Paper 2

Further practice · Attempt part b.ii only.

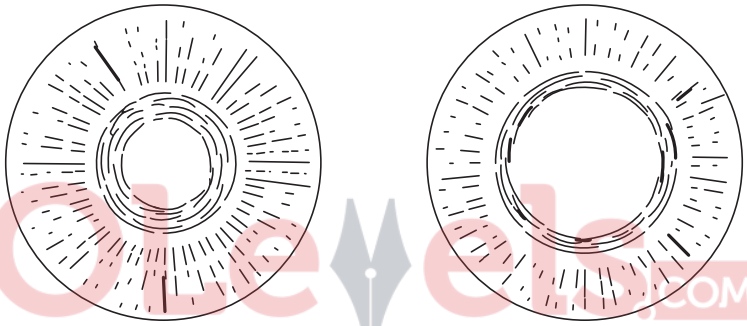
6 The human eye consists of many parts.

(a) Which part has the highest concentration of light sensitive cells?

(1)

- ☒ **A** blind spot
- ☒ **B** choroid
- ☒ **C** fovea
- ☒ **D** optic nerve

(b) The diagram shows the iris of an eye in two different light conditions, A and B.



light condition A light condition B

(i) Describe what happens in the eye when the light condition changes from A to B.

(5)

DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA

2. Question 6 · part b.ii · 3 marks

June 2021 Paper 2

Further practice · Attempt part b.ii only.

DO NOT WRITE IN THIS AREA

(ii) Explain why it is important that the changes in the eye occur quickly when the light condition changes from B to A.

(3)

(Total for Question 6 = 9 marks)

OLevels.com SCHOOL

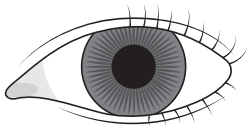
Working and answer

3. Question 6 · part c.i · 3 marks

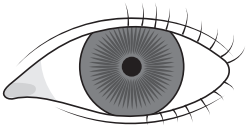
June 2024 Paper 2

Further practice · Attempt part c.i only.

6 The diagram shows how the size of the pupil changes in different light intensities.



Dim light



Bright light

(a) Describe a method that could be used to investigate how the size of the pupil changes in different light intensities.

(3)

(b) A student calculated the area of the pupil in bright light to be 7.1 mm^2.

Using information from the diagram, calculate the area of the pupil in dim light.

[for a circle, area = πr^2 $\pi = 3.14$]

(3)

area of pupil in dim light = ... mm^2

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

P 7 4 6 3 0 A 0 1 6 2 4

3. Question 6 · part c.i · 3 marks

June 2024 Paper 2

Further practice · Attempt part c.i only.

DO NOT WRITE IN THIS AREA

(c) (i) Explain how the size of the pupil is adjusted from bright light to dim light.

(3)

(ii) Explain the adjustment made to the pupil when moving into dim light.

(2)

(Total for Question 6 = 11 marks)

OLevels.com SCHOOL

3. Question 6 · part c.i · 3 marks · Working and answer

Working and answer

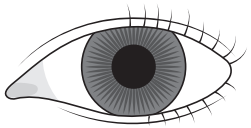
OLevels.com SCHOOL

4. Question 6 · part c.ii · 2 marks

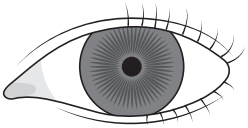
June 2024 Paper 2

Further practice · Attempt part c.ii only.

6 The diagram shows how the size of the pupil changes in different light intensities.



Dim light



Bright light

(a) Describe a method that could be used to investigate how the size of the pupil changes in different light intensities.

(3)

(b) A student calculated the area of the pupil in bright light to be 7.1 mm^2.

Using information from the diagram, calculate the area of the pupil in dim light.

[for a circle, area = πr^2 $\pi = 3.14$]

(3)

area of pupil in dim light = ... mm^2

DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA

OLevels.com SCHOOL

4. Question 6 · part c.ii · 2 marks

June 2024 Paper 2

Further practice · Attempt part c.ii only.

DO NOT WRITE IN THIS AREA

(c) (i) Explain how the size of the pupil is adjusted from bright light to dim light.

(3)

(ii) Explain the adjustment made to the pupil when moving into dim light.

(2)

(Total for Question 6 = 11 marks)

DO NOT WRITE IN THIS AREA

OLevels.com SCHOOL

Working and answer